MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING, TYPE 5902WA

This specification is approved for use by the Naval Electronics Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Beam power, subminiature

Outline	---	3-3(EIA)
Base	---	E8-10 (Button base)
Envelope	---	T-3
Cathode	---	Coated unipotential

Base connections:

Pin no.	---	1	2	3	4	5	6	7	8
Element	---	g1	k g3	h	k g3	a	h	g2	k g3

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	eb	Ec1	Ec2	Rk	Rg1
Unit:	V	V dc	v	V dc	V dc	Ohms	Meg
Maximum:	6.6	250	360	0	155	---	0.55
Minimum:	6.0	---	---	-55	---	---	---
TEST CONDITIONS:	6.3	110	---	0	110	270	---

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ehk	Pp	Pg2	Ik	TE	Alt
Unit:	V dc	W	W	mA dc	°C	ft
Maximum:	±200	3.7	0.4	50	220	(see note 1)
Minimum:	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---	---

GENERAL:

Qualification - Required

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Insp level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 1								
1301	Heater current			0.4	II	If	427	473	ma
1336	Heater-cathode leakage			0.4	II	Ihk	---	10	µA dc
1266	Total grid current		Rg1 = 1.0 Meg	0.4	II	Ic	0	-1.0	µA dc
1256	Electrode current (1) (anode)			0.4	II	Ib(1)	24	36	mA dc
1256	Electrode current (2) (anode)		Ec1 = -40 V dc; Rk = 0	0.4	II	Ib(2)	---	100	µA dc
1341	Power output		Esig = 6.4 V ac; Rp = 3,000 ohms	0.4	II	Po	0.75	---	W
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
3011	Insulation of electrodes			2.5	I	R	250	---	Meg
3306	Transconductance			2.5	I	Sm	3,500	4,900	µmhos
1256	Electrode current (screen grid)			2.5	I	Ic2	0	4.0	mA dc
1266	Grid emission	2	Ef = 7.5 V; Eb = 250 V dc; Ec2 = 150 V dc; Ec1 = -40 V dc; Rk = 1,350 ohms; Rg1 = 1.0 Meg	2.5	I	Isc1	0	-2.0	µA dc
1341	Change in power output of individual tubes (1)		Ef = 5.7 V; Rp = 3,000 ohms; Esig = 6.4 V ac	2.5	I	ΔPo Ef	---	15	%
1246	Audio frequency noise	3	Esig = 150 mV ac; Ec1 = -8.7 V dc; Rg1 = 0.5 Meg; RL = 2,000 ohms; Rk = 0; Cg2 = 4 µf; Ecc2 = 110 V dc; Rg2 = 10,000 ohms	2.5	I	---	---	---	---
3131	Direct-interelectrode capacitance		0.405 in. dia. shield	6.5	Code F	Cgp	---	0.20	pF
			0.405 in. dia. shield			Cin	5.5	7.5	pF
			0.405 in. dia. shield			Cout	6.5	8.5	pF

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Insp level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 2 -Continued								
1311	Anode resistance			6.5	S4	rp	0.01	---	Meg
1256	Electrode current (1) (anode) (tube operation time)	4		4.0	S4	t	---	20	sec
1031	High-frequency vibration		Rp = 2,000 ohms	2.5	I	Ep	---	30	mVac
1031	Sweep-frequency vibration		F = 70 to 2,000 Hz; 15 G; Rp = 2,000 ohms; t = 3 minutes; positions X1 and X2 only	6.5	Code G	ep	---	200	mvpp
1116	Lead fatigue			2.5	Code F	---	4.0	---	arcs
1041	Shock	5	450 G; Ehk = +100 V dc	6.5	Note 6	---	---	---	---
1031	Sweep-frequency-vibration fatigue		F = 130 to 2,000 Hz	6.5	Note 6	---	---	---	---
---	Shock and sweep-frequency-vibration fatigue-test end points								
1031	High-frequency vibration		Rp = 2,000 ohms	---	---	Ep	---	120	mVac
1336	Heater-cathode leakage			---	---	Ihk	---	20	µA dc
1341	Change in power output of individual tubes (2)		Esig = 6.4 V ac; Rp = 3,000 ohms	---	---	ΔPo t	---	20	%
1266	Total grid current		Rgl = 1.0 Meg	---	---	Ic	0	-3.0	µA dc
2126	Envelope strain			2.5	I	---	---	---	---

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.0 V; Eb = Ec2 = 0; Ehk = 140 V ac; 1 min "on", 4 min "off"	---	---	---	---
---	Heater-cycling life-test end point						
1336	Heater-cathode leakage			Ihk	---	15	µA dc
1516	Stability life	7	Eb = 250 V dc; Ec2 = 150 V dc; Rk = 1,350 ohms; Ehk = +200 V dc; Rgl = 0.47 Meg; TA = Room	---	---	---	---
---	Stability life-test end point						
1341	Change in power output of individual tubes (2)	7	Esig = 6.4 V ac; Rp = 3,000 ohms	ΔPot	---	15	%
1501	Intermittent life (1)	8,9	Eb = 250 V dc; Ec2 = 150 V dc; Rk = 1,350 ohms; Ehk = +200 V dc; Rgl = 0.47 Meg; TE = 200°C (min)	---	---	---	---
---	Intermittent life (1) - test end points (500 hours)	10					
1266	Total grid current		Rgl = 1.0 Meg	Ic	0	-2.0	µA dc
1301	Heater current			If	414	492	mA
1341	Change in power output in individual tubes (2)		Esig = 6.4 V ac; Rp = 3,000 ohms	ΔPot	---	20	%
1341	Change in power output in individual tubes (1)		Ef = 5.7 V; Esig = 6.4 V ac; Rp = 3,000 ohms	ΔPo Ef	---	15	%
1336	Heater-cathode leakage			Ihk	---	20	µA dc
1211	Insulation of electrodes			R	50	---	Meg
1501	Intermittent life (2)	8,9	Eb = 100 V dc; Ec2 = 100 V dc; Rk = 220 ohms; Ehk = +200 V dc; Rgl = 0.47 Meg; TE = 220°C (min)	---	---	---	---

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3 -Continued						
---	Intermittent life (2) - test end points (500 hours)	10					
1266	Total grid current		Rgl = 1.0 Meg	Ic	0	-2.0	µA dc
1301	Heater current			If	414	492	mA
1341	Change in power output of individual tubes (2)		Esig = 6.4 V ac; Rp = 3,000 ohms	ΔPo t	---	20	%
1341	Change in power output of individual tubes (1)		Ef = 5.7 V; Esig = 6.4 V ac; Rp = 3,000 ohms	ΔPo Ef	---	15	%
1336	Heater-cathode leakage			Ihk	---	20	µA dc
1211	Insulation of electrodes			R	50	---	Meg
1341	Average change in power output of individual tubes (2)		Esig = 6.4 V ac; Rp = 3,000 ohms	Avg ΔPo t	---	15	%
---	Intermittent life (2) - test end points (1000 hours)	10					
1266	Total grid current		Rgl = 1.0 Meg	Ic	0	-2.0	µA dc
1301	Heater current			If	414	492	mA
1341	Change in power output of individual tubes (2)		Esig = 6.4 V ac; Rp = 3,000 ohms	ΔPo t	---	25	%
1341	Change in power output of individual tubes (1)		Ef = 5.7 V; Esig = 6.4 V ac; Rp = 3,000 ohms	ΔPo Ef	---	20	%
1336	Heater-cathode leakage			Ihk	---	20	µA dc
1211	Insulation of electrodes			R	25	---	Meg

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.

2. Prior to this test, tubes shall be preheated a minimum of 5 minutes at the conditions indicated below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef	Eb	Ec1	Ec2	Rg	Rk
V	V dc	V dc	V dc	Meg	ohms
7.5	250	0	150	1.0	1350

3. The rejection level shall be set at the VU meter reading obtained during calibration.

4. Tube operation time is the time in seconds required for the anode current to attain 85 percent of the 2-minute anode current value, when measured at the test conditions specified for electrode current (1) (anode) test. The total filament supply impedance of the transformer and series external resistance shall be 1.75 ohms ±20 percent. The applied filament voltage shall be adjusted so that the heater voltage listed under test conditions is the actual voltage applied to the tube under test terminals for a bogey tube or equivalent.

5. A grid resistor of 0.1 megohm shall be added, except when a thyratron-type short indicator is used.

6. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of a lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

7. The AQL shall be 1.0 percent defective, inspection code I.

8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT, having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position on the life-test rack.

9. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples and shall not exceed a total of three tube failures.

10. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples.

11. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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